

WAVE MOTION

→ The transfer of disturbance or energy from one region of medium to another by the successive vibration of particles of the medium is called wave or wave motion.

> Types of Wave

① Mechanical wave

→ The wave which requires medium for its propagation is called mechanical wave.
 e.g.: Sound wave

The properties of medium required for the propagation of mechanical wave are:-

- ① Elasticity
- ② Inertia

② Electromagnetic wave

→ The wave which do not require medium for its propagation is called electromagnetic wave.
 e.g.: Light

③ Matter wave or de-Broglie wave

→ The wave associated with moving particle is called Matter wave or de-Broglie wave.
 e.g.: Electron wave

> Types of Mechanical wave (on the basis of vibration of particles of medium)

① Longitudinal Wave

④ → The wave in which the particles of medium vibrate in the direction of wave propagation is called Longitudinal wave.
E.g: Sound wave in air.

↳ During the propagation of wave, alternate compressions and rarefaction are formed, at compression, Pressure is high and at rarefaction, Pressure is low.

That means there is Variation in the pressure of medium when longitudinal wave propagates through it. So, longitudinal wave is also called pressure wave.

For the propagation of Longitudinal wave, bulk modulus of elasticity of medium is required. All the media i.e. solid, liquid and gas possesses bulk modulus of elasticity. So, longitudinal wave propagate through solid, liquid and gas.

② Transverse wave

↳ The wave in which particle of medium vibrate perpendicular to the direction of wave propagation. is called Transverse wave. E.g: wave in stretched string.

During the propagation of Transverse wave, Alternate crests and troughs are formed.

For the propagation of transverse wave, the medium should possess modulus of rigidity. Solid possess modulus of rigidity. So transverse wave can propagate through solid but ^{not} through liquid and gas. Due to surface tension, the free surface of liquid maintains its level. Transverse wave can also be formed in the liquid surface.

• Wave Parameters

① Displacement (y)

→ The distance of vibrating particle from its mean position is called displacement.

② Amplitude (A)

→ The maximum displacement of vibrating particle from the mean position is called amplitude.

③ Time period (T)

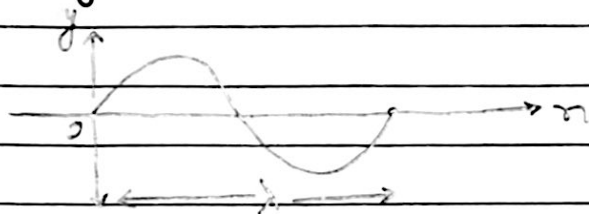
→ The time taken by the particle to complete one oscillation is called time period.

④ Wave Frequency (f)

→ The number of waves formed per second is called wave frequency.

⑤ Wavelength (λ)

- It is the distance travelled by wave during one complete vibration of particle of the medium.
- It is the distance between two nearest particles of the medium which are vibrating in same phase.



⑥ Wave velocity or phase velocity

- The distance travelled by wave per unit time is called wave velocity or phase velocity.

$$V = \frac{\text{distance travelled}}{\text{time}} = \frac{\lambda}{T} = f \times \lambda$$

Wave velocity is also given by

$$V = \frac{\omega}{k}$$

⑦ Particle velocity

- During the propagation of a wave through a medium, the particles of the medium are in simple harmonic motion. So, particle velocity is given by

$$V = \omega \sqrt{x^2 - y^2}$$

Where x is the amplitude and y is the

displacement of the particle from its mean position.

③ Angular wave number or propagation constant

↳ It is the change in phase per unit path difference.

$$\text{i.e. } k = \frac{d\phi}{dx} = \frac{2\pi}{\lambda}$$

• Phase

↳ The state of oscillating particle at a given time is called phase of the particle. Phase of a particle at any instant refers to the position and direction of oscillation of the particle at that instant. It is expressed in terms of angle.

→ Relation between phase difference and path difference.

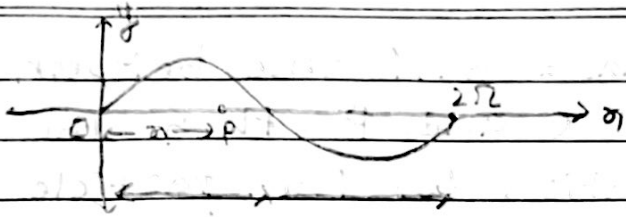
For the path difference λ , phase difference
= 2π rad

For the path difference 1 , phase difference
= $\frac{2\pi}{\lambda}$ rad.

For the path difference n , phase difference
= $\frac{2\pi}{\lambda} \times n$

∴ phase difference (ϕ) = $\frac{2\pi}{\lambda} \times \text{path difference } (n)$

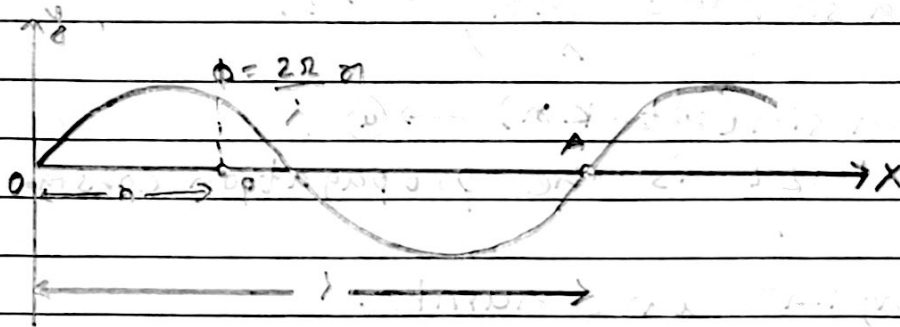
$$v = \frac{\omega}{k}$$



• Progressive wave

→ The wave which travels in Forward direction with definite velocity is called progressive wave.

> Progressive wave equation



Consider a wave is travelling along positive x -direction. during the propagation of wave, the particles of medium execute SHM. so, the displacement equation for the particle at O is given by.

$$y = a \sin \omega t \rightarrow (i)$$

Where a is the amplitude

Let us take another particle at P . The particle at P starts oscillation after the particle at O . so, the phase of particle at P lags behind the particle at O .

Let ϕ be the phase difference between the particles at O and P. Then the displacement equation for the particle at P is given by

$$y = a \sin(\omega t - \phi) \rightarrow (2)$$

For opposite direction, $y = a \sin(\omega t + \phi)$

If x be the path difference between the particles at O and P then

$$\phi = \frac{2\pi}{\lambda} x$$

Then, equation (2) can be written as,

$$y = a \sin\left(\omega t - \frac{2\pi x}{\lambda}\right) \rightarrow (3)$$

$$\therefore y = a \sin(\omega t - kx) \rightarrow (4)$$

where $k = \frac{2\pi}{\lambda}$ is the propagation constant

or angular wave number.

Since $\omega = \frac{2\pi}{T}$ then from (3)

$$y = a \sin\left(\frac{2\pi}{T} t - \frac{2\pi x}{\lambda}\right)$$

$$\therefore y = a \sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right) \rightarrow (5)$$

Again, $\omega = 2\pi f = \frac{2\pi v}{\lambda}$ then from (3)

$$y = a \sin\left[\frac{2\pi v}{\lambda} t - \frac{2\pi x}{\lambda}\right]$$

$$\therefore y = a \sin \frac{2\pi}{\lambda} [vt - x] \rightarrow (6)$$

The above equations are the different forms of progressive wave equation.

Q. The plane progressive wave is represented by $y = 0.1 \sin\left(200\pi t - \frac{17\pi x}{20}\right)$ where

y is in mm, t is in sec and x is in meter. find

(i) amplitude, frequency, wavelength and velocity of wave.

(ii) The maximum particle velocity.

(iii) The wave equation having double the amplitude but travelling in opposite direction.

Ans.

Here, $y = 0.1 \sin\left(200\pi t - \frac{17\pi x}{20}\right)$

(i) $a = ?$, $f = ?$, $\lambda = ?$, $v = ?$

Comparing given eqⁿ with $y = a \sin(\omega t - kx)$ we have:

$$\therefore a = 0.1 \text{ mm} = 0.1 \times 10^{-3} \text{ m}$$

$$\omega = 200\pi$$

$$2\pi f = 200\pi$$

$$\therefore f = 100 \text{ Hz}$$

Now,

$$k = \frac{17\pi}{20}$$

$$v = f \times \lambda$$

$$= 100 \times \frac{40}{17}$$

$$\frac{2\pi}{\lambda} = \frac{17\pi}{20}$$

$$= \frac{4000}{17}$$

$$\therefore \lambda = \frac{40}{17} = 2.35 \text{ m}$$

$$= 235.3 \text{ m/s}$$

(ii) $v_{\text{max}} = ?$

We know that,

$$v_{\text{max}} = \omega a$$

$$= 200 \times 0.1 \times 10^{-3}$$

$$= 0.02 \text{ m/s}$$

$$(iii) y = 2 \times 0.1 \sin \left(200\pi t + \frac{17\pi}{20} \right)$$

$$\therefore y = 0.02 \sin \left(200\pi t + \frac{17\pi}{20} \right)$$

• Principle of superposition

↳ It states that when a number of waves travelling through a medium meet at a point then resultant displacement of the waves is given by the vector sum of displacements due to individual waves.

If $\vec{y}_1, \vec{y}_2, \dots, \vec{y}_n$ be the displacements due to n waves then resultant displacement when all the waves act together is given by

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \dots + \vec{y}_n$$

• Stationary wave or standing wave

↳ The resultant wave formed by the superposition of two progressive waves having same amplitude and same frequency but

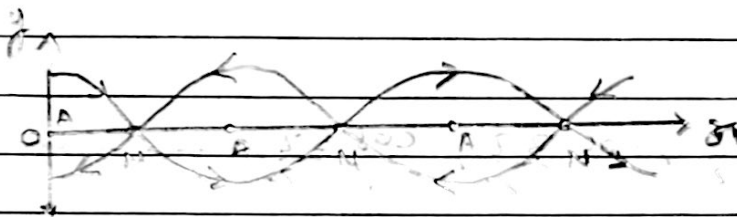
travelling in opposite directions is called a stationary wave.

Consider two progressive waves of same amplitude 'a' and same frequency 'f' but travelling opposite directions are.

$$y_1 = a \sin(\omega t - kx) \rightarrow (i)$$

$$\text{And } y_2 = a \sin(\omega t + kx) \rightarrow (ii)$$

Where ω is the angular frequency and k is the angular wave number.



When these two waves superposed to each other then a stationary wave is formed. According to the principle of superposition, the displacement of the resultant wave, i.e. stationary wave is given by

$$y = y_1 + y_2$$

$$= a \sin(\omega t - kx) + a \sin(\omega t + kx)$$

$$= a [\sin(\omega t - kx) + \sin(\omega t + kx)]$$

$$= a \times 2 \sin\left(\frac{\omega t - kx + \omega t + kx}{2}\right) \cos\left(\frac{\omega t - kx - \omega t - kx}{2}\right)$$

$$\cos\left(\frac{\omega t - kx - \omega t - kx}{2}\right)$$

$$= 2a \sin \omega t \cos(-kx).$$

$$= 2a \cos kn \cdot \sin \omega t \quad [\because \cos(-A) = \cos A]$$

$$= A \sin \omega t \rightarrow \text{vii)}$$

Where, $A = 2a \cos kn \rightarrow \text{viii)}$

Here, eqn vii) represents the stationary wave eqn and eqn viii) gives amplitude of the stationary wave.

> Position of nodes

At the nodes, amplitude is zero.

i.e. $A = 0$

or, $2a \cos kn = 0$

or, $\cos kn = 0$

or, $\cos kn = \cos \frac{\pi}{2}, \cos \frac{3\pi}{2}, \cos \frac{5\pi}{2}, \dots$

or, $\cos kn = \cos (2n-1) \frac{\pi}{2}$ Where $n = 1, 2, 3, \dots$

or, $kn = (2n-1) \frac{\pi}{2}$

or, $\frac{2\pi}{\lambda} n = (2n-1) \frac{\pi}{2}$

or, $n = (2n-1) \frac{\lambda}{4} \rightarrow \text{ix)}$

i.e. $n = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$ are the positions

of nodes.

Now, distance between any two consecutive nodes $= \frac{3\lambda}{4} - \frac{\lambda}{4}$ (or $\frac{5\lambda}{4} - \frac{3\lambda}{4}$) & so on.

$$= \frac{\lambda}{2}$$

→ At the Antinode, amplitude is Maximum.

~~is Ant~~

Antinodes are the points in the stationary wave at which the displacement is maximum.

Amplitude (A) is maximum ~~which~~ at antinodes, which is possible if

$$\cos kn = \pm 1$$

$$\text{or, } \cos kn = \cos 0, \cos \pi, \cos 2\pi, \cos 3\pi, \dots$$

$$\text{or, } \cos kn = \cos n\pi \quad \text{where } n = 0, 1, 2, 3, \dots$$

$$\text{or, } kn = n\pi$$

$$\text{or, } \frac{2\pi}{\lambda} n = n\pi$$

$$\text{or, } n = \frac{n\lambda}{2} \rightarrow (v)$$

i.e. $n = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots$ etc are the pos-

ition of ~~amplitude~~ antinodes.

Now, distance between two consecutive antinodes = $\frac{\lambda}{2} - 0$ (or, $\lambda - \frac{\lambda}{2}$ and so on)

$$= \frac{\lambda}{2}$$

Again, the distance between two consecutive node and antinode

$$= \frac{\lambda}{4} - 0 \quad \left(\text{or } \frac{3\lambda}{4} - \frac{\lambda}{2} \text{ and so on} \right)$$

$$= \frac{\lambda}{4}$$

Conclusion,

The distance between two consecutive nodes or two consecutive antinodes in a stationary wave is $\frac{\lambda}{2}$ and the distance between two consecutive node and antinode is $\frac{\lambda}{4}$.

- Differences between progressive and stationary wave

Progressive Wave	Stationary wave
① Energy travels forward with definite velocity.	Energy cannot transfer from one region of medium to another.
② Each particles of the medium execute SHM with the same amplitude.	Except nodes, all particles of the medium execute SHM with varying amplitude.
③ No particle of the medium is permanently at rest.	The particles of the medium at nodes are permanently at rest.
④ There is continuous change of phase from one particle to the next.	All the particle between two successive nodes vibrate in the same phase.

• Wave Properties

- ① Wave undergo reflection, refraction, interference and diffraction.
- ② In reflection and refraction, frequency of the wave remain unchanged. so, frequency is considered as fundamental quantity in wave.
- ③ The phase difference of π takes place when a wave is reflected from a denser medium.
- ④ No phase difference takes place when a wave is reflected from a rarer medium.
- ⑤ In refraction, no phase difference takes place.
- ⑥ Transverse wave can be polarised but longitudinal wave cannot be polarised.

Conceptual Questions.

- ① Do sound waves under go reflection, refraction and polarization phenomena? Explain.
- Sound waves show reflection and refraction, but not polarization. Reflection occurs as echoes when sound bounces off a surface;
Refraction occurs when sound bends on entering a medium where its speed changes. &
Polarization does not occur because sound is a longitudinal wave - particles vibrate along the direction of propagation, so there is no transverse plane of vibration to restrict.

- ② A radio station broadcasts at 800 kHz . What will be the wavelength of the wave?

→ Soln,

$$f = 800 \times 10^3 \text{ Hz}$$

$$\lambda = ?$$

We know that,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{800 \times 10^3} = \frac{3 \times 10^8}{8 \times 10^5} = 375 \text{ m}$$

- ③ Can longitudinal wave be polarized? Explain

→ No, a longitudinal wave cannot be polarized.

Polarization confines vibration to one plane perpendicular to wave travel - possible only in transverse waves. In longitudinal waves, particles vibrate parallel to the direction of travel, so there is no plane to confine.

- ④ Explain what is meant by the frequency of a wave.

→ Frequency is the number of complete oscillations (cycles) of a wave passing a fixed point per second. Unit: Hertz (Hz).

- ⑤ Ultrasound is a longitudinal wave. Describe the nature of a longitudinal wave.

→ A longitudinal wave has particle vibrations parallel to the direction of wave propagation, producing alternating compression and rarefactions that transmit energy through the medium.

Numerical

- ① In order to produce an image with sufficient detail, the wavelength of the ultrasound must be 0.50 mm. The speed of the ultrasound in body tissue is 1540 m/s. Calculate the frequency of the ultrasound at this wavelength.

↳ Soln,

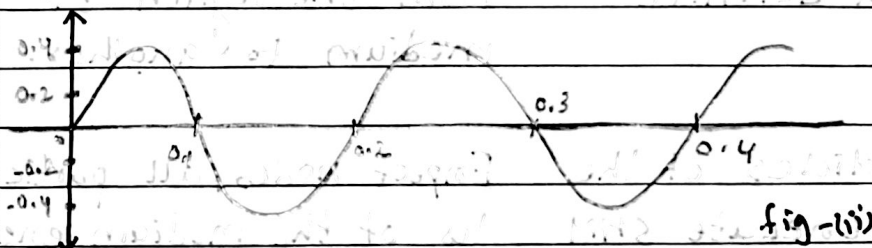
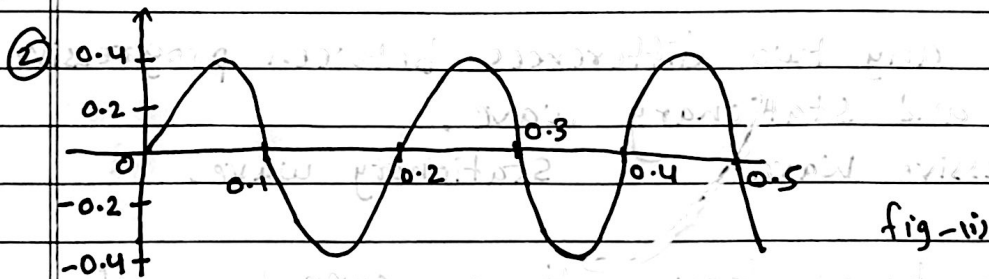
$$\lambda = 0.50 \times 10^{-3} \text{ m}$$

$$v = 1540 \text{ m/s}$$

$$f = ?$$

We know that,

$$f = \frac{v}{\lambda} = \frac{1540}{0.50 \times 10^{-3}}$$
$$= 3.08 \times 10^6 \text{ Hz}$$



Use fig-(i) and (ii) to determine.

a) the amplitude

→ amplitude = $0.4 \text{ mm} = 0.4 \times 10^{-3} \text{ m}$

b) Wavelength = 10 cm

= 0.1 m

c) frequency (f) = ?

$T = 0.2 \times 10^{-3} \text{ s}$

$f = \frac{1}{T} = \frac{1}{0.2 \times 10^{-3}}$

= 5000 Hz

d) The speed.

$V = f \times \lambda$

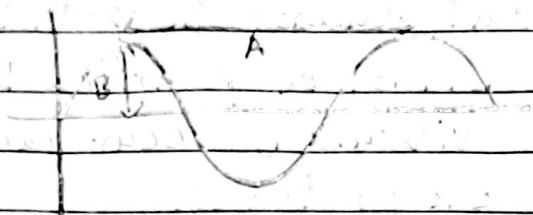
= 5000×0.1

= 500 m/s

e) Write any two differences between progressive wave and stationary wave.

Progressive Wave	Stationary wave
① Energy travels forward with definite velocity.	Energy cannot transfer from one region of medium to another.
② Each particles of the medium execute SHM with the same amplitude.	Expect nodes, all particles of the medium execute SHM with varying amplitude.

③ The diagram shows a transverse wave.



i) Name the distances labelled A and B. Also, define them.

→ A = Wavelength

∴ Distance travelled by wave in one complete oscillation is wavelength.

B = Amplitude

∴ Maximum displacement of vibrating particle from the mean position is called amplitude.

ii) 20 waves pass a fixed point every second. What is frequency of the wave?

→ Soln,

$$\begin{aligned} \text{frequency} &= \text{Total waves pass per second} \\ &= \frac{20}{1} \\ &= 20 \end{aligned}$$

$$\therefore \text{frequency} = 20 \text{ Hz}$$

iii) Calculate the velocity of the wave if distance A = 1.5 m.

→ Soln,

$$\cancel{V = f \times \lambda} \quad V = f \times \lambda$$

$$\begin{aligned} V &= 20 \times 1.5 \quad [\because A = \lambda = 1.5 \text{ m}] \\ &= 30 \text{ m/s} \end{aligned}$$

iv) Transverse Waves can be polarised. Name a type of wave that cannot be polarised.

→ Longitudinal wave cannot be polarised because their particle vibrations occur parallel to the direction of energy travel.

Exam pattern Questions.

2.

d) A small piece of cork in a ripple tank oscillates up and down simple harmonically as ripple passes. If the ripple travel at 0.20 m/s have a wavelength of 15 mm and an amplitude of 5 mm, what is the max. velocity of cork.

→ Soln,

$$V = 0.20 \text{ m/s}$$

$$\lambda = 15 \times 10^{-3} \text{ m}$$

$$a = 5 \times 10^{-3} \text{ m}$$

$$V_{\text{max}} = ?$$

We know that.

$$V_{\text{max}} = \omega a$$

$$= \frac{2\pi}{\lambda} a$$

$$= \frac{2\pi \times 0.20}{15 \times 10^{-3}} \times 5 \times 10^{-3}$$

$$= \frac{2\pi \times 0.20}{15} \times 5$$

$$= 2\pi \times 0.20$$

$$= 2\pi \times \frac{V}{\lambda} a$$

$$= \frac{2\pi \times 0.20}{15 \times 10^{-3}} \times 5 \times 10^{-3}$$

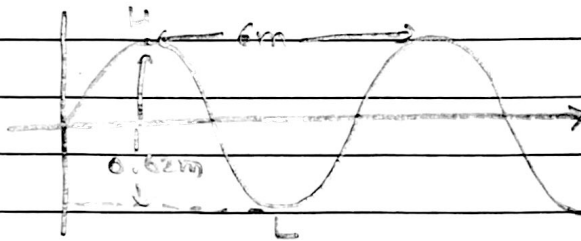
$$= \frac{2\pi \times 0.20}{3}$$

$$= 0.42 \text{ m/s}$$

3.

c) A fisherman notices that his boat is moving up and down periodically owing to waves on the surface of water.

It takes 2.5 sec for the boat to travel from its highest point to its lowest, a total distance of 0.62 m. The fisherman sees that the wave crests are spaced 6 m apart as shown in figure.



i) Amplitude = ?

$$\text{Amplitude (a)} = \frac{0.62}{2}$$

$$= 0.31$$

ii) How fast are the wave travelling.

c) Soln,

$$\lambda = 6 \text{ m}$$

$$a = 0.62 \text{ m}$$

$$t = 2.5 \times 2 = 5 \text{ s}$$

Now,

$$v = f \times \lambda$$

$$= \frac{1}{T} \times \lambda$$

$$= \frac{1}{5} \times 6$$

$$= 1.2 \text{ m/s.}$$

iii) The phase difference between two consecutive troughs is 2π rad.